

Task 2

If the expected PnL is positive, the client is good/profitable for the LP. If the expected PnL is negative, the client is costly for the LP.

Clients A, B, and C look profitable. Their PnL values stay positive across all horizons. Client A looks the best, followed by Client B and then Client C. Since their aggregate PnL is already positive, they do not need any extra minimum half-spread to make the PnL non-negative.

Client D is more mixed. It makes positive PnL at shorter horizons, but the PnL starts getting weaker at longer horizons. Its aggregate PnL is still slightly positive, so I would classify it as marginally profitable, but not as safe as A, B, and C.

Clients E and F are costly for the LP. Their PnL becomes negative, especially at longer horizons. Client F looks the worst because it has the most negative aggregate PnL. This means the LP would need to charge a higher spread for Client F to avoid losses.

The minimum half-spread $\delta^*\delta^*$ is connected to Task 1 adversity. Clients with higher adversity percentages usually have worse PnL. So, more adverse clients need a larger $\delta^*\delta^*$. This matches the Task 1 result, where Client A was the safest and Client F was the most adverse.

Assumptions:

I calculated everything from the LP perspective. Aggregate PnL is calculated using uniform closing weights as given in Eq. 6. I classified clients with aggregate PnL ≥ 0 as profitable and clients with aggregate PnL < 0 as costly.